15. Estimate the de Broglie wavelength of a baseball thrown by a pitcher.

de Broglie wavelength: $\lambda = \frac{h}{p} = \frac{h}{mv}$

$$= \frac{(6.626 \times 10^{-34} \frac{kg \cdot m^2}{s})}{(0.145 kg)(45 m/s)} \approx 1.0155 \times 10^{-34} m$$

1. We said that continuity required $\psi(L) = 0$ and that led us to conclude that $\sqrt{2mE}\,L/\hbar$ had to equal $n\pi$. Each of the questions below has a different answer: a different reason why we didn't include certain values of n.

(a) Why didn't we include non-integer values of n among our solutions?

(b) Why didn't we include $n = 0$?

(c) Why didn't we include negative (integer) values of n?

a.) this is due to the principles of quantization which arises from the conditions of our boundaries where $\sin(n\pi) = 0$

b.) If $n=0$ were true then the particle simply would not exist $\sin(0) = 0$

c.) negative integer values suggest the shape and phase of our wavefunction would change and since probability $|\psi|^2$ (this can't be the case)

11. Equations (5.7) and (5.8) represent the solution to Schrödinger's equation in an infinite square well of length L. Write down the solution for an infinite square well of length $2L$. *Hint*: If this takes you more than 30 seconds, you're working too hard.

The sine function gives you a zero at $\pm\pi$, $\pm 2\pi$, etc.:

$$\frac{\sqrt{2mE}}{\hbar}L = n\pi \quad \rightarrow \quad E = \frac{\hbar^2\pi^2}{2mL^2}n^2, \quad \text{where } n \text{ is any positive integer.} \tag{5.7}$$

$$\psi(x) = \begin{cases} 0 & x \le 0 \\ \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi}{L}x\right) & 0 < x < L \\ 0 & x \ge L \end{cases} \quad \text{Energy eigenstates of the infinite square well.} \tag{5.8}$$

5.7

$$E = \frac{\hbar^2\pi^2}{8mL^2}n^2$$

5.8

$$\psi(x) \begin{cases} 0 & x \le 0 \\ \sqrt{\frac{1}{L}}\sin\left(\frac{n\pi x}{2L}\right) & 0 \le x \le 2L \\ 0 & x \ge 2L \end{cases}$$

12. Consider the infinite square well (Figure 5.4) with one change: on the domain $0 < x < L$, instead of $U = 0$, the potential energy is $U = U_0$ (a constant). (You examined this case qualitatively in Question 3; now you're going to do the math.)

(a) Write down Schrödinger's equation for a particle in this field on the domain $0 < x < L$.

(b) Rearrange your solution and write down the general solution, including two new constants A and B. (We haven't given you any "technique" for doing this, but the equation is similar enough to the one we solved that you should be able to work something out. Don't stop until you have mathematically confirmed that your $\psi(x)$ function works in your differential equation!)

(c) The boundary condition $\psi(0) = 0$ immediately tells you the value of one of your arbitrary constants. Apply that and write down a simpler $\psi(x)$ function than you had before.

(d) Now apply the boundary condition $\psi(L) = 0$ to limit the possible values of the constant E. Remember that U_0 is a constant!

(e) How is your solution like, and unlike, the solution we found for $U = 0$? Briefly explain the significance of your result.

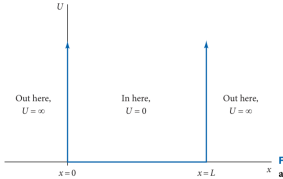

Figure 5.4 Potential energy $U(x)$ for an "infinite square well."

a.) $\mathcal{E}\psi(x) = -\frac{\hbar}{2m}\frac{d^2}{dx}\psi(x) + U(x)\psi(x)$

$\psi = \psi(x)e^{-i\omega t}$

$\mathcal{E} = i\hbar\frac{d}{dt} = \underbrace{\hbar\omega}_{\mathcal{E}}\psi$

guess $\psi(x) \propto e^{ikx}$

$\hookrightarrow \mathcal{E}\cancel{\psi} = \frac{-\hbar^2k^2\cancel{\psi}}{2m} + U(x)\cancel{\psi}$

$\hookrightarrow \mathcal{E} = \frac{-\hbar^2k^2}{2m} + U_0$

B.) · Find A by normalization

* total probability must equal 100% *

$\boxed{1} = \int dx\,|\psi(x)|^2 = \int_0^L dx\, A^2\sin^2\left(\frac{\pi x}{L}\right)dx$

$\psi(x) = A\sin kx + B\cos kx$

$\lambda = \frac{2L}{N}$

try: $\sin^2\theta = \frac{1-\cos 2\theta}{2}$: $\int ax\cos bx = \frac{1}{b}\sin bx$

$\rightarrow A^2\left(\frac{L}{2} - \frac{L}{\pi}\cancel{\sin\frac{\pi x}{L}}^{\,0}\Big|_0^L\right) \Rightarrow \boxed{1} = A^2\frac{L}{2} \rightarrow A = \sqrt{2/L}$, $\sin(kL) = 0 \Rightarrow k = \frac{n\pi}{L}$

c/d.)

$\Rightarrow$ boundary @ $x=0$: $\left[e^{ik0} + e^{-ik0}\right] = b_3$

$\psi_2(x) = Ae^{ikx} + Be^{-ikx}$ $\quad \psi_1(0) = \psi_2(0)$

$\psi_1(x) = a_1e^{kx} + b_1e^{-kx}$: $\psi_2(L) = \psi_3(L)$

$\psi_3(x) = a_3e^{kx} + b_3e^{-kx}$

$\left.\frac{d\psi_1}{dx}\right|_{x=0} = \left.\frac{d\psi_2}{dx}\right|_{x=0}$

$\left.\frac{d\psi_3}{dx}\right|_{x=L} = \left.\frac{d\psi_2}{dx}\right|_{x=L}$

$\mathcal{E} = \frac{-\hbar^2k^2}{2m} + U_0$

$\left(\frac{\hbar^2k^2}{2m}\right) = \mathcal{E} - U \Rightarrow k^2 = \frac{2m(E-U_0)}{\hbar^2}$, $E - U_0 = \frac{\hbar^2k^2}{2m} = \frac{\hbar^2}{2m}\left(\frac{n\pi}{L}\right)^2$, $\mathcal{E} = U_0 + \frac{\hbar^2\pi^2n^2}{2mL^2}$

e.) It is similar except for the $U(x)$ term which we can limit to the constant U_0

16. A particle has the potential energy function $U(x) = U_0$ for $x \le 0$, $U(x) = 0$ for $0 < x < L$, and $U(x) = 2U_0$ for $x \ge L$ (where $U_0 > 0$).

(a) Write Schrödinger's equation in each of these three regions.

(b) Write the general solution to Schrödinger's equation in each region, assuming $0 < E < U_0$. Your answer should have a total of six arbitrary constants. Use only real formulas, so make sure your square roots are all of positive quantities.

(c) Which two of those arbitrary constants must equal 0 and why?

(d) Figure 5.11 shows an energy eigenstate for a symmetric finite square well. How do the eigenstates of this problem's *asymmetric* well differ from that one, and what does that difference imply about the probability distribution for position?

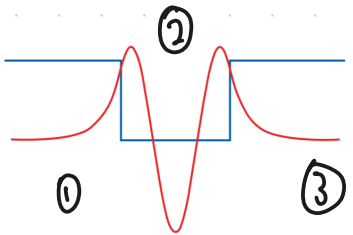

Figure 5.11 An energy eigenstate (red) of a finite square well (potential energy in blue).

$$U(x) = \begin{cases} U_0 & x \le 0 \\ 0 & 0 \le x \le L \\ 2U_0 & x \ge L \end{cases} \quad (U_0 > 0)$$

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + U(x)\psi = E\psi$$

$$\hookrightarrow \frac{d^2\psi}{dx^2} = \frac{2m}{\hbar^2}(U-E)\psi$$

a.) ① $\frac{d^2\psi}{2m} = \frac{2m}{\hbar}(U_0 - E)\psi$

where $K_1 = \sqrt{\frac{2m(U_0-E)}{\hbar}} > 0 \Rightarrow \frac{d^2\psi}{dx^2} = K_1^2\psi$

② $\frac{d^2\psi}{2m} = \frac{2m}{\hbar}(2U_0 - E)\psi$

where $K_2 = \sqrt{\frac{2m(E)}{\hbar}} > 0 \Rightarrow \frac{d^2\psi}{dx^2} = -K_2^2\psi$

③ $\frac{d^2\psi}{2m} = \frac{2m}{\hbar}(2U_0 - E)\psi$

where $K_3 = \sqrt{\frac{2m(2U_0-E)}{\hbar}} > 0 \Rightarrow \frac{d^2\psi}{dx^2} = K_3^2\psi$

b.)

$\psi_1(x) = Ae^{K_1 x} + Be^{-K_1 x}$

as $x \to \infty$, $\psi_1(x) = Ae^{K_1 x} + \underline{0}$

$\psi_2(x) = a_2 \sin(kx) + b_2 \cos(kx)$

$\psi_3(x) = a_3 e^{K_3 x} + b_3 e^{-K_3 x}$

as $x \to \infty$, $\psi_3(x) = b_3 e^{K_3 x}$

c.) B and a_3 must be zero, because as $x \to \infty$ $e^{K_1 x}$ and $e^{-K_3 x} \to \infty$ and are non-normalizable

17. A Half Finite Well

A particle has the potential energy function $U(x) = \infty$ for $x \leq 0$, $U(x) = 0$ for $0 < x < L$, and $U(x) = U_0$ for $x \geq L$.

(a) What is $\psi(x)$ in the region $x \leq 0$?

(b) Write the general solution to the Schrödinger equation using real functions in each of the two remaining regions, assuming $0 < E < U_0$. Your answer should have a total of four arbitrary constants.

(c) Two of the arbitrary constants from Part (b) must be zero. Which ones, and why?

(d) Write an equation relating the two remaining arbitrary constants, and explain how you know it is true. Use it to rewrite the wavefunction in all of space with only one arbitrary constant.

(e) Write an equation that the energy eigenvalue E must satisfy. Your equation should include m, L, U_0, and $\hbar$, but not any of the arbitrary constants. (You do not need to solve this equation.)

a.) If $U=\infty$, the particle cannot exist there, so: $\psi(x)=0$ w/ condition $U(0)=0$

b.) @ $0 \leq x \leq L$, $U=0$

General

so: $\frac{d^2\psi}{dx^2} = -k^2\psi$, $k=\sqrt{\frac{2mE}{\hbar^2}}$ thus: $\psi_2(x) = A\sin(kx) + B\cos(kx)$

$\psi_2(0) = A\sin 0 + B\cos 0 = B = 0$ so: $\psi_2(x) = A\sin(kx)$

@ $x \geq L$, $U = U_0$

so: $\frac{d^2\psi}{dx^2} = K^2\psi$, $K=\sqrt{\frac{2m(U_0-E)}{\hbar^2}}$ thus: $\psi_3(x) = Ce^{Kx} + De^{-Kx}$

and as $x \to \infty$ we find: $C=0$, so: $\psi_3 = De^{-Kx}$

c.) $B=0$ and $C=0$

from $\psi(0)=0$

normalizing as $x \to \infty$

d.) to satisfy E, we must find $\psi_2(L) = \psi_3(L)$: $A\sin(kL) = De^{-KL}$, $D = A\sin(kL)e^{KL}$

where $\frac{d\psi_2}{dx} = Ak\cos(kx) \Rightarrow \frac{d\psi_2}{dx} = Ak\cos(kL)$

$\frac{d\psi_3}{dx} = -KDe^{-Kx} \Rightarrow \frac{d\psi_3}{dx} = -KDe^{-KL}$

$Ak\cos(kL) = -KDe^{-KL}$

and for K: $K = -k\tan(kL)$

$\hookrightarrow \tan(kL) = \frac{-K}{k}$